

新乡市弘力电源科技有限公司

圆柱形锂离子可充电电池

产品规格书

XINXIANG HONGLI SUPPLY SOURCE TECHNOLOGY CO., LTD

SPECIFICATIONS

CYLINDRICAL LITHIUM-ION RECHARGEABLE BATTERY

型 号 MODEL : IMR18650-1200mAh

客户名称 CUSTOMER:

客户确认 CONFIRMATION:

日 期 DATE : 2025-08-01

制定 PRERARED	审核 CHECKED	批准 APPROVED
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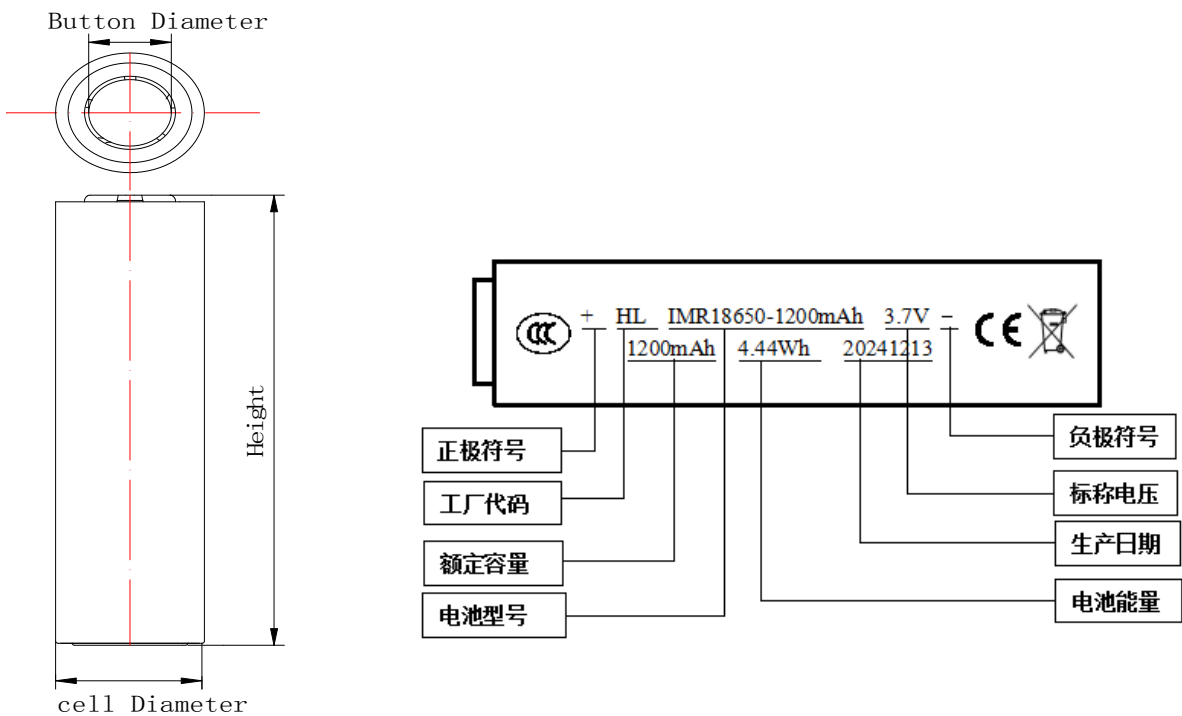
1.适用范围 Scope

本产品规格书描述了新乡市弘力电源科技有限公司（以下简称弘力公司）生产的圆柱锂离子电池的产品性能指标。

The document describes the product specification of the cylindrical lithium-ion rechargeable battery cell supplied by Xinxiang Hongli supply source technology co., LTD（Hereinafter referred to as Hongli company）

2.型号 Model: IMR18650-1200mAh

3.包装尺寸及喷码:



Cell diameter(电池直径)	Cell Height(电池高度)	Button diameter(盖帽直径)
mm	mm	mm
18.2±0.3	65.0±0.5	7.5±0.2(平)
18.2±0.3	67.3±0.5	6.0±0.2(尖)

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4.规格

序号NO.	项目 Item	规格 Specification
1.	标称容量 Nominal capacity	1200mAh@ 0.2C（240mA）放电discharge
2.	额定能量 Rated energy	4.44Wh
3.	标称电压 Nominal voltage	3.7V
4.	内阻 Impedance	≤60mΩ（充电态 1 KHZ ）
5.	充电限制电压 limited charging voltage	4.20V
6.	充电上限电压 upper limited charging voltage	4.20V
7.	标准充电电流 Standard Charging Current	240mA
8.	最大充电电流 Maximum charging current	600mA
9.	标准充电方法 Standard Charging method	0.2C ₅ A 恒流（CC）充电至 4.2V，再恒压 4.2V（CV）充电直至充电电流≤0.02C ₅ A 0.2C ₅ A constant current(CC)charge to 4.2V, then consrant voltage 4.2V（CV）charge till charge current decline to≤0.02C ₅ A
10.	充电时间 Charging time	标准充电：7.5 h（参考值）Standard charge: 7.5 h（reference） 快速充电：3.5 h（参考值）Rapid charge: 3.5 h（reference）
11.	标准放电电流 Standard discharging current	240mA
12.	最大放电电流 Maximum discharging current	600mA
13.	放电终止电压 end of discharge voltage	3.0V
14.	放电截止电压 Discharge cut-off voltage	2.75V
15.	上限充电温度 upper limited charging temperature	45℃
16.	下限充电温度 lower limited charging temperature	0℃

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17.	上限放电温度 upper limited discharging temperature	60℃						
18.	下限放电温度 lower limited discharging temperature	-10℃						
19.	储存温度 Storage temperature	-10℃ ~ 45℃						
20.	储存湿度 Storage humidity	≤85%RH						
21.	电池重量 Cell Weight	约Approx: 33.9g						
22.	容量保持	搁置时间	1个月	2个月	3个月	6个月	9个月	12个月
		容量保持率	92%	91%	90%	86%	83%	80%

5.电池性能 Battery Performance

5.1 电性能 Electrical characteristics

序号 NO.	项目 Item	标准 Requirements	测试方法 Test Method
1	常温放电性能 Discharge Characteristics	A) 0.2C ₅ A ≥5h B) 0.5C ₅ A ≥115min C) 1C ₅ A ≥54min	在 1 标准大气压,环境温度 23℃±2℃,相对湿度为 45%~75%的条件下, 电池标准充电后（以下若没有特别说明, 均在此条件下放置, 皆按此充电方式）, 搁置 30min, 分别以 0.2C ₅ A、0.5C ₅ A、1C ₅ A、进行放电至终止电压, 循环三次, 当有一次达到标准, 即达到标准要求（下同）。 Standard charged under the condition of normal atmospheric pressure and the environmental temperature of 23℃±2℃ and 45%~75%RH, then rest for 30min and discharge at 0.2C ₅ A、0.5C ₅ A、1C ₅ A to the discharge cut-off voltage respectively. Charge/discharge cycle can be conducted for 3 times before meeting the Standards (the same below) .
2	常温荷电保持能力 Norm Storage	放电时间≥4.25h Discharge Time≥4.25h	在 1 标准大气压,环境温度 23℃±2℃, 电池标准充电后, 开路放置 28 天, 再以 0.2C ₅ A 放电至终止电压。 Under conditions of 1 standard atmosphere pressure and ambient temperature of 23℃±2℃, after standard charging of the battery, it is left open-circuit for 28 days, then discharged to the termination voltage at 0.2C ₅ A.

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3	循环寿命 Cycle Life	Cdischarge ≥ 150 times	在 1 标准大气压, 环境温度 23℃±2℃, 相对湿度为 45%~75%的条件下, 电池标准充电后, 搁置 30min, 标准放电至终止电压, 循环至标称容量 80%循环次数。 Under conditions of 1 standard atmosphere pressure, environmental temperature of 23℃±2℃, and relative humidity of 45% to 75%, after standard charging of the battery, it is left to stand for 30 minutes, then discharged to the termination voltage under standard conditions, and the number of cycles until the nominal capacity reaches 80% is determined.
4	出货电压 Cell Voltage	出货状态 As of shipment	3.80V-4.05V

5.2 环境适应性能 Adaptation to Environment Characteristic

序号NO.	项目 Item	标准 Standard	测试方法 Test Method
1	热循环性能 Thermal Cycle	电池不漏液、不冒烟、不起火、不爆炸 No smoking 、No fire No explosion、No leakge	电池标准充电后, 在环境温度为 75±2℃的条件下开路放置 48h, 在-20℃条件下开路放置 6h, 然后在室温条件下开路放置 24h, 1C ₅ A 进行放电至终止电压。 Standard charge the battery, then store it at 75±2℃ for 48h, then -20℃ for 6h and room temperature for 24h. Then discharge at 1C ₅ A to discharge cut-off voltage.
2	恒定湿热性能 Static Humidity	电池不起火、不爆炸 放电时间≥36min No explosion、No fire Discharge Time ≥36min	电池标准充电后, 置于温度为 40℃±2℃, 相对湿度为 90%的恒温恒湿箱中, 搁置 48h 后, 取出电池 23±2℃环境下搁置 2h。观察电池外观变化。然后以 1C ₅ A 放电至终止电压。 Standard charge. Put the battery into a 40℃±2℃ and 90% RH chamber for 48h, then get it out and store it for 2h at room temperature. Observe the variation of the battery's appearance and then discharge at 1C ₅ A to discharge cut-off voltage, measuring final capacity.
3	跌落性能 Drop	不起火、不爆炸 No fire、No explosion	标准充电后, 将电池样品由高度(最低点高度)为 1m 的位置自由跌落到 18~20mm 的混凝土板上, 要求各面各跌落一次。 Standard charge. Then let it self fall off from a height of 1.0m(the lowest height) to a smooth wooden surface. The self fall off should be

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			conducted from every positive and negative direction.
4	不同温度下的 放电性能 High-low Temperature Discharge	电池不爆炸、不起火 No fire\explosion A)55 °C ≥90% B)0 °C ≥80%; C)-10 °C ≥60%	电池标准充电后，在 55±2℃条件下恒温搁置 3h、以 0.2C₅A 放电至终止电压，然后在室温条件下标准充电，依次按照 0±2 °C/-10±2 °C 的顺序在相应的恒温条件下搁置 20h，以 0.2C₅A 放电至终止电压。 Standard charge. Then store for 3h at 55±2 °C and discharge at 0.2C₅A to discharge cut-off voltage. then standard charge at room temperature and store for 20h according to the order of 0±2 °C/-10±2 °C and discharge at 0.2C₅A measuring corresponding discharge capacity.
5	振动环境 适应性能 Vibration	电池外观无明显损伤、不漏液、不冒烟、不爆炸 No remarkable damage 、 No smoking 、 No explosion、 No leakge	电池标准充电后，安装在振动台面上，按下面的振动频率和对应的振幅调整好试验设备,X、Y、Z 三个方向每个方向上从 10Hz~55Hz 循环扫频振动 30min，扫频速率为 1oct/min: A)振动频率: 10Hz~30Hz 位移幅值（单振幅）: 0.38mm B)振动频率: 30Hz~55Hz 位移幅值（单振幅）: 0.19mm。 Standard charge. Measure initial status. Equip it to the vibration platform, adjust and prepare the test equipment according to following vibration frequency and relevant swing, doing frequency sweeping from X, Y, Z three directions, each from 10Hz to 55Hz for 30 minutes of recycling, rating of which is 1oct/min: A)vibration frequency:10Hz~30Hz Displacement breadth (single swing): 0.38mm B) vibration frequency: 30Hz~55Hz Displacemen tbreadth(single swing): 0.19mm。 Measure final status after sweeping and Observe the variation of the battery's appearance.

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6	热冲击测试 Heating Test	不起火、不爆炸 No fire, no explosion	测量电池的初始状态，电池标准充电后，放置于热箱中，并与热电偶相连，温度以（5℃±2℃）/min 的速率升至 130℃±2℃并保温 10min。观察电池外观变化。 Measure the initial state of the battery, after standard charging, place it in a hot box, , And connected to a thermocouple, the temperature rises at a rate of (5℃± 2℃)/min 130℃± 2℃ and insulation for 10 minutes. Observing changes in battery appearance.
7	常温短路测试 Room temperature short circuit test	不爆炸、不起火 No explosion, no fire	电池标准充电后，测量电池的初始状态，置于防爆玻璃罩中直接短路其正负极（线路总电阻 80±20mΩ），当电池温度下降到比峰值约低 10℃ 时试验结束观察电池的温度及外观变化。 After standard charging of the battery, measure the initial state of the battery and place it in explosion-proof glass Short circuit its positive and negative poles directly in the cover (the total resistance of the circuit is 80±20mΩ), when The test ends when the battery temperature drops to about 10℃ below the peak value. Observing electricity The temperature and appearance changes of the pool.
8	高温短路测试 High temperature short circuit test	不爆炸、不起火 No explosion, no fire	电池充电后，测量电池的初始状态，放置在 57℃±4℃的环境中，待电池表面温度达到 57℃±4℃后，再放置 30 min。然后在此环境温度下用导线连接电池正负极端，并确保全部外部电阻为 80±20mΩ。试验过程中监测电池温度变化，电池温度下降值达到温度最大值的 20%，或短接时间达到 24 h，实验终止。 After charging the battery, measure its initial state, and place it in an environment of 57℃± 4℃. Wait until the surface temperature of the battery reaches 57℃± 4℃, and then keep it for another 30 minutes. Then, at this environmental temperature, connect the positive and negative terminals of the battery with wires and ensure that the total external resistance is 80mΩ ± 20mΩ. During the test, monitor the temperature changes of the battery. The test will be terminated when the temperature drop of the battery reaches 20% of the maximum temperature or when the short-circuit time reaches 24 hours.
9	重物冲击 Impact Test	不起火、不爆炸 No fire, no explosion	电池标准充电后，测量电池的初始状态，放在平面上，并与热电偶相连，将一直径为 15.8mm 的棒放在电池高度方向中间位置，让重量 9.1kg 的重物从 610mm 高度自由垂落至冲击台面，观察电池外观及温度变化。 After standard charging of the battery, measure the initial state of the battery, place it on a flat surface, and connect it to a thermocouple. Place a rod with a diameter of 15.8mm in the middle of the height direction of the battery, allowing a weight of 9.1kg to

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			freely fall from a height of 610mm to the impact table, and observe the appearance and temperature changes of the battery.
10	挤压测试 Crush Test	不起火、不爆炸 No fire, no explosion	测量电池的初始状态，电池标准充电后，与热电偶相连，放置于两铁制平面模具中，用 13KN 的最大力压力进行瞬间压缩，观察电池的温度及外观变化。 Measure the initial state of the battery, connect it to a thermocouple after standard charging, and place it in two iron flat molds. Apply a maximum force of 13KN for instant compression, and observe the temperature and appearance changes of the battery.
11	过放电测试 Overdischarge test	不起火、不爆炸 No fire, no explosion	将电芯放完电，再用 1C 反接充电 90 分钟 A discharged cell is subjected to a reverse charge at 1C for 90 min.
12	过充电测试 Overcharge test	不爆炸、不起火 No fire, no explosion	电芯标准充满电后，以 4.6V 的恒定电压继续充电，保持 8 小时 After standard charge, continue to charge with a constant voltage 4.6V per a cell, holding 8h.

备注：以上标准中的一些术语的定义：

Commts: The definitions of some nomenclatures of this specification.

(1) 标准充电：在环境温度 23℃±2℃ 的条件下，以 0.2C₅A 充电，当电池端电压达到充电限制电压 4.2V 时，改为恒压充电，直到充电电流小于或等于 0.02C₅A 后停止充电

standard charge: Charge with current 0.2C₅A to limit charge voltage 4.2V under the condition of 23℃±2℃ surrounding temperature, then change to charge with constant voltage till the current less than or equal to 0.02C₅A.

(2) 剩余容量：电池经过特定的检测程序后的首次放电容量。

Residual Capacity: The first discharge capacity after being tested by the specific procedure.

(3) 恢复容量：电池经过特定的检测程序后，通过反复充放电使状态恢复后的放电容量。

Recovery Capacity: The discharge capacity by implementing charge-discharge cycle repeatedly after being tested by the specific procedure.

(4) 用于上述测试的电池必须是交货一个月内的电池，除非另有规定。

Test should be conducted with new batteries within one month after shipment from our factory, Unless otherwise defined.

6.外观检查 Visual Inspection

不允许有任何影响电池性能的外观缺陷，如裂纹、裂缝、泄漏等。

There shall be no such defect as scratch, flaw, crack, and leakage, which may adversely affect commercial value of the cell.

7.标准测试环境 Standard environmental test condition

除非特别说明，本规格书中所有测试均在以下环境条件下进行：

Unless otherwise specified, all tests stated in this Product Specification are conducted at below condition:

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温度: $23 \pm 2^{\circ}\text{C}$ Temperature: $23 \pm 2^{\circ}\text{C}$
湿度: 25-85% RH Humidity: 25-85% RH
大气压: 86KPa~106 kPa Atmosphere: 86KPa~106 kPa

8. 储存及其它事项 Storage and Others

8.1 长期储存 Long Time Storage

长期储存的电池（超过3个月）须置于干燥、凉爽处。每6个月对电池进行一次充放电，储存电压为3.6~3.9V且储存环境要求如7。电池在储存或周转过程中，盒中的油膜纸禁止去除，以防电池生锈。

If the cell is stored for a long time(exceed three months), the cell should be stored in drying and cooling place. The cell should be charged and discharged every six month. The cell's storage voltage should be 3.6~3.9V and the cell is to be stored in a condition as NO.7. During storage or handling, the oil paper in the box must not be removed to prevent the cell from rusting.

8.2 其它事项 Others

任何本规格书中未提及的事项，须经双方协商确定。

Any matters that this specification does not cover should be consulted between the customer and HUANYU.

9. 保质期及产品责任 Warranty Period& Product Liability

保质期是从出厂日期（喷码）开始起 6 个月

Warranty period of this product is 6 months from manufacturing code.

弘力公司对因没有按本规格书规定操作而导致的意外不负责任，当本规格书有一些变动时，弘力公司会通知购买方。

HONGLI company is not responsible for the troubles caused by mishandling of the battery which is clearly against the instructions in this specification. When HONGLI company find any new facts which require modification of this document, we will inform the buyer.

10. 电池使用时警告事项及注意事项 Warnings and Cautions in using the battery

为防止电池可能发生泄漏,发热、爆炸,请注意以下预防措施:

To prevent the possibility of the battery from leaking, heating or explosion please observe the following precautions:

警 告 ! WARNINGS!

- 使用规定的充电器
- Use the battery charger specifically for that purpose when recharging
- 不要将电池投入火中或加热
- Do not discard the battery in fire or a heater.
- 不要将电池分解拆散
- Do not dismantle the battery
- 严禁将电池浸入海水或水中,保存不用时,应放置于阴凉干燥的环境中.
- Do not immerse the battery in water or seawater, and keep the battery in a cool dry surrounding if it stands by.
- 禁止将电池在热高温源旁,如火、加热器等使用和留置.
- Do not use or leave the battery near a heat source as fire or heater.
- 充电时请选用锂离子电池专用充电器.
- Please choose lithium-ion battery charger when charging.
- 严禁颠倒正负极使用电池.
- Do not reverse the position and negative terminals.
- 严禁将电池直接插入电源插座.
- Do not connect the battery to an electrical outlet.
- 禁止用金属直接连接电池正负极短路.

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- Do not short-circuit the battery by directly connecting the positive and negative terminals with metal objects.
- 禁止将电池与金属,如发夹、项链等一起运输或贮存.
- Do not transport or store the battery together with metal objects such as hairpins, necklaces, etc.
- 禁止敲击或抛掷、踩踏电池等.
- Do not strike, trample or throw the battery.
- 禁止直接焊接电池和用钉子或其它利器刺穿电池.
- Do not directly solder the battery and pierce the battery with a nail or other sharp objects.

小心 ! CAUTIONS!

- 禁止在高温下（炙热的阳光下或很热的汽车中）使用或放置电池,否则可能会引起电池过热、起火或功能失效、寿命减短.

Do not use or leave the battery at high temperature (for example, at strong direct sunlight or in a vehicle in extremely hot weather). Otherwise, it can overheat or fire or its performance will be degenerate and its service life will be decreased.

- 禁止在强静电和强磁场的地方使用,否则易破坏电池安全保护装置,带来不安全的隐患.
Do not use the battery in a location where static electricity and magnetic field is great, otherwise, the safety devices may be damaged, causing hidden trouble of safety.
- 如果电池发生泄露,电解液进入眼睛,请不要揉擦,应用清水冲洗眼睛,并立即送医治疗,否则会伤害眼睛.

If the battery leaks, and the electrolyte get into the eyes, do not rub the eyes, instead, rinse the eyes with clean water, and immediately seek medical attention. Otherwise, it may injure eyes.

如果电池发出异味,发热、变色、变形或使用、贮存、充电过程中出现任何异常,立即将电池从装置或充电器中移离并停用.

If the battery gives off an odor, generates heat, becomes discolored or deformed, or in any way appear abnormal during use, recharging or storage, immediately remove it from the device or battery charger and stop using it.

如果电极弄脏,使用前应用干布抹净,否则可能会导致接触不良功能失效

In case the battery terminals are dirty, clean the terminals with a dry cloth before use. Otherwise performance may occur due to the poor connection with the instrument.

- 废弃之电池应用绝缘纸包住电极,以防起火、爆炸。
- Be aware discarded batteries may cause fire or explosion, tape the battery terminals to insulate them.